Read each of the questions carefully and answer showing all steps of your solution

Final Answers with no clear solution get ZERO

Q1- Find the root of the function: $f(x) = Sin(x) + x^2 + x - 10$ using: *(12 marks)*

(a) The ***Bisection Method***. Perform two iterations to find the root, use the interval *(2, 4)*

x_{r3} = ...

(b) The ***Newton Raphson Method***. Perform two iterations to find the root, assume $x_0 = 2$.

x_2 = ...

Note: work to 5 significant figures

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lve the following systems of linear equations **(12 marks)**

a) Use the ***Cramer's Rule*** to solve the following two simultaneous equations, determine x_1 and x_2:

$$x_1 + x_2 = 25$$

$$6x_1 - 3x_2 = 6$$

x_1 = ..., x_2 = ...

(b) Use the ***Gauss Elimination Method*** to solve the following three simultaneous equations. Determine the values of x, y, z:

$$-3x - y + 2z = 25$$

$$4x + 2y + z = 16$$

$$5x + 3y - \frac{z}{2} = 3$$

x = ..., y = ..., z = ...

Q3- Use the ***Bairstow Method*** applying first iteration, to find the roots of the polynomial: **(6 marks)**

$f(x) = x^3 - 2x^2 + x - 2$. Use $(x^2 - 0.5x + 1)$, as a starting factor.

Given the results of the first synthetic division are:

$$b_3 = 1, b_2 = -1.5, b_1 = -0.75, b_0 = -0.875$$

End of Questions

$\Rightarrow$ (Q2.2)

[redacted]

(a.) $\Rightarrow$ sol: $f(X) = \sin(x) + x^2 + x - 10$ / $X_l = 2$, $X_u = 4$

Zero ~~[illegible]~~ iteration

$\Rightarrow X_{r_0} = \frac{2+٤}{2} = 3$ / $f(X_l) = -3.0907$ / $f(X_u) = 9.2٤32$

$f(X_r) = 2.1٤11$

الآلة
Rad

$\Rightarrow$ new interval $\rightarrow X_u = 3$ / $X_l = 2$

~~{[illegible] iteration}~~ {2 st iteration}

$X_{r_1} = \frac{3+2}{2} = 2.5$ / $f(X_l) = -3.0907$ / $f(X_u) = 2.1٤11$
$f(X_r) = -0.6515$

2^{nd} $\Rightarrow$ new interval $\rightarrow X_l = 2.5$ / $X_u = 3$

~~[illegible]~~ iteration

$\Rightarrow X_{r_2} = \frac{2.5+3}{2} = \boxed{2.75}$

Amazing

ⓑ $\Rightarrow$ sol:

$$f(x) = \sin(x) + x^2 + x - 10 \; / \; x_0 = 2$$

$$f'(x) = \cos(x) + 2x - 1$$

الآلة
Rad

& $x_{i+1} = x_0 - \frac{f(x_0)}{f'(x_0)}$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \Rightarrow x_1 = 2 - \frac{-3.0907}{2.5838}$$

$$\boxed{x_1 = 3.1962}$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \Rightarrow x_2 = 3.1962 - \frac{3.3573}{4.3939}$$

$$\boxed{x_2 = 2.4321}$$

[Q.2]

ⓐ $x_1 + x_2 = 25$

$6x_1 - 3x_2 = 6$

=> solv

$$[A] = \begin{bmatrix} 1 & 1 \\ 6 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 25 \\ 6 \end{bmatrix}$$

=> To find [X] :- => $\det\{A\} = (1 \times -3) - (1 \times 6)$

$= -3 - 6 = -9$

$$[A_x] = \begin{bmatrix} 25 & 1 \\ 6 & -3 \end{bmatrix} \Rightarrow X = \frac{\det\{A_x\}}{\det\{A\}}$$

$\det(A_x) = (25 \times -3) - (1 \times 6)$

$= -81$

$$X = \frac{-81}{9} = -9 \Rightarrow \boxed{X = -9}$$

=> To find (y) :-

$$[A_y] = \begin{bmatrix} 1 & 25 \\ 6 & 6 \end{bmatrix} \rightarrow \det(A_y) = (1 \times 6) - (25 \times 6)$$

$= -144$

$$\Rightarrow y = \frac{\det(A_y)}{\det(A)} = \frac{-144}{-9} = 16$$

Amazing

$\boxed{y = 16}$ //

$\Rightarrow$ (b) sol:

$$-3x - y + 2z = 25$$
$$4x + 2y + z = 16$$
$$5x + 3y - \frac{z}{2} = 3$$

$$\Rightarrow \left[\begin{array}{ccc|c} -3 & -1 & 2 & 25 \\ 4 & 2 & 1 & 16 \\ 5 & 3 & -0.5 & 3 \end{array}\right] \quad 4R_1 + 3R_2$$

$$\left[\begin{array}{ccc|c} -3 & -1 & 2 & 25 \\ 0 & 2 & 11 & 148 \\ 5 & 3 & -0.5 & 3 \end{array}\right] \quad 5R_1 + 3R_3 \longrightarrow \left[\begin{array}{ccc|c} -3 & -1 & 2 & 25 \\ 0 & 2 & 11 & 148 \\ 0 & 4 & 8.5 & 134 \end{array}\right]$$

$-2R_2 + R_3$

$$\left[\begin{array}{ccc|c} -3 & -1 & 2 & 25 \\ 0 & 2 & 11 & 148 \\ 0 & 0 & -13.5 & -162 \end{array}\right]$$

$$0(x) + 0(y) - 13.5z = -162$$
$$\Rightarrow \boxed{z = 12}$$
$$\Rightarrow 0(x) + 2(y) + (11 \times 12) = 148$$
$$\boxed{\Rightarrow y = 8}$$
$$\Rightarrow -3(x) - 8 - (2 \times 12) = 25 \Rightarrow \boxed{x = -19}$$

starting factor

$\Rightarrow f(x) = x^3 - 2x^2 + x - 2 \quad / \quad x^2 - 0.5x + 1$

[redacted] → دائماً ثابتين [redacted]

$\Rightarrow p = -0.5 \quad / \quad q = +1$

⇐ هسا هو مدبحني من أول قسمة لأنه معطيني معاملات ⇐ $(\underline{b})$

⇐ باخذهم وبستخدمهم بالقسمة

١

	1	-1.5	-0.75	-0.875
$-p = 0.5$		0.5	-0.5	
$-q = -1$			-1	
	1	-1	-2.25	$\boxed{}$
	c_3	c_2	c_1	

بس بالقسمة [redacted] معكوس إشاراتهم

→ بدخل فاضية لأني ما بحتاج

$\boxed{C_0}$ → بالمعادلات

٢

$$\Rightarrow c_2 \delta p + c_3 \delta q = b_1 \quad --- ①$$

$$\Rightarrow (c_1 - b_1)\delta p + c_2 \delta q = b_0 \quad --- ②$$

$\Rightarrow -\delta p + \delta q = -0.75 \quad \text{---} ①$

$\Rightarrow (-2.25 \overset{(--)}{+} 0.75)\, \delta p - \delta q = -0.875 \quad \text{--} ②$

نحصل على الآلة الحاسبة معادلات بمجهولين

مقدار خطأ δp (X)

مقدار خطأ δq (Y)

$\Rightarrow \delta p = 0.65 \;/\; \delta q = -0.1$

ج

$\Rightarrow x^2 + (p + \delta p)\, x + (q + \delta q)$

$\Rightarrow x^2 + (\bar{0}.5 + 0.65)\, x + (1 + (\bar{0}.1))$

$\Rightarrow x^2 + 0.15\, x + 0.9 = 0$

$\Rightarrow x = \underline{\underline{-0.075}} \pm \underline{\underline{0.9457}}\, \hat{i}$

ل (α) ل (B)